

ConcreteAI v4.0

Concrete Mix Proportion Designer

User Manual

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1. Application Overview

ConcreteAI v4.0 is an AI-assisted concrete mix proportion design application developed at the Concrete Technology Laboratory, Faculty of Engineering, KMUTT. The application integrates ACI 211.1 standard proportioning procedures with a trained Multi-Layer Perceptron (MLP) neural network model to predict compressive strength (f_c) at 28 days.

The tool targets civil engineers, researchers, and graduate students who require rapid, reliable mix design guidance across multiple concrete types. All calculations are performed client-side via the React front-end with optional connection to a FastAPI backend for AI-enhanced predictions.

1.1 Supported Concrete Types

Type	Full Name	f_c Range	Workability
NC	Normal Concrete	15 – 60 MPa	Slump 50–200 mm
HPC	High Performance Concrete	60 – 95 MPa	Flow 150–750 mm
SCC	Self-Compacting Concrete	20 – 100 MPa	Flow 450–750 mm
UHPC	Ultra-High Performance Concrete	100 – 150 MPa	Flow 150–400 mm
HSLWC	High-Strength Lightweight Concrete	20 – 55 MPa	Slump 50–180 mm
GEO	Geopolymer / Alkali-Activated	5 – 55 MPa	Slump 50–200 mm

Note: GEO type is available in Research Mode only and requires specification of alkaline activator parameters (NaOH molarity, SS/SH ratio).

2. System Requirements

2.1 Front-End (Standalone Mode)

- Modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- JavaScript must be enabled
- Minimum screen width: 375 px (mobile-first responsive design)
- Internet connection for Google Fonts (optional; falls back to system fonts)

2.2 Back-End (AI-Enhanced Mode)

- Python 3.9 or higher
- FastAPI + Uvicorn
- scikit-learn, pandas, numpy
- pyngrok (for Google Colab tunnel)
- Back-end URL configurable via REACT_APP_API_BASE environment variable

Important

The application operates in two independent modes: (1) Local ACI Mode when the API is unavailable, providing ACI 211.1-based proportions without AI prediction; and (2) AI-Enhanced Mode when connected to the FastAPI backend, providing MLP-predicted f_c values with 95% confidence intervals.

3. Getting Started

3.1 Launching the Application

Open the application in a supported browser. A welcome guide overlay will appear automatically on first load. Navigate through all four guide slides using the 'Next' button, or close the guide with the X button in the top-right corner of the overlay. The guide can be re-opened at any time using the '?' button in the application header.

3.2 Interface Layout

Zone	Location	Purpose
Header	Top of screen	Institutional logos, app title, API status indicator, help button
Mode Bar	Below header	Switch between Normal Mode, Research Mode, and History tab
Input Panel	Scrollable body	Concrete type selector, f'c input, workability slider, binder configuration
Results Panel	Appears on calc	Mix proportions grid, AI prediction box, carbon footprint breakdown
History Tab	Third tab	Saved calculations, compare tool, delete and reload functions

4. Normal Mode — Step-by-Step

Normal Mode is designed for practising engineers who require a rapid mix design estimate with minimal inputs. GEO (Geopolymer) is not available in this mode.

Step 1 — Select Concrete Type

Tap or click the desired concrete type card from the list. The card displays the type name, Thai sub-label, MPa range, and workability type. The selected card highlights with an orange border and a tick mark.

Step 2 — Enter Target Compressive Strength

Type the target 28-day compressive strength (f'_c) in the numeric input field. The unit can be toggled between MPa and ksc using the toggle buttons on the right. An inline hint shows the allowable range for the selected concrete type. An error state (red border) is triggered if the value is out of range; the Calculate button will be disabled until corrected.

Step 3 — Adjust Workability

Drag the slider to set the target Slump (mm) or Flow (mm). The label above the slider automatically reflects the correct workability measure for the selected type. HPC uses Flow only; the toggle between Slump and Flow is available for SCC and UHPC.

Step 4 — Optional Secondary Binder

A secondary supplementary cementitious material (SCM) can be added in Normal Mode for NC, HPC, SCC, UHPC, and GEO types. Select one of: FA (Fly Ash), BA (Bottom Ash), BFS (GGBS), SF (Silica Fume), CC (Calcium Carbonate), or 'None'. When a secondary binder is selected, a ratio slider (5%–40%) appears. The primary binder proportion is automatically reduced to accommodate the chosen percentage.

Step 5 — Calculate

Press the 'Calculate Mix Proportion' button. A loading spinner appears while the API is queried. Results are displayed immediately and automatically saved to the History tab.

5. Research Mode — Step-by-Step

Research Mode exposes full binder composition control and is the only mode where Geopolymer concrete (GEO) is available.

5.1 Binder Configuration

- Up to 10 binder materials can be combined simultaneously.
- The first binder (primary) is pre-filled based on the selected concrete type.
- Tap '+ Add Binder' to open the picker and select additional materials from the available list.
- Each binder row shows the material name, full IUPAC/common name, a percentage input field, and a remove button.
- A running total badge shows the current sum; the Calculate button is disabled until the sum reaches exactly 100%.

5.2 GEO — Alkaline Activator Parameters

When GEO is selected in Research Mode, an additional 'Alkaline Activator' panel appears beneath the binder configuration.

Parameter	Range	Default	Description
NaOH Molarity	4 – 16 mol/L	10	Sodium hydroxide solution concentration
SS / SH Ratio	0.5 – 4	2.5	Sodium silicate to sodium hydroxide mass ratio
Activator / Binder	0.30 – 0.60	0.45	Total activator to binder mass ratio

6. Interpreting Results

6.1 Mix Proportion Summary

All quantities are expressed in kg per cubic metre (kg/m³) of fresh concrete. The result header bar shows the selected type, approximate unit weight, CO₂ emissions, and the calculated Water/Binder (W/B) ratio.

6.2 AI Prediction Box

The AI Prediction Box displays:

- Predicted f'_c at 28 days (MPa) — point estimate from the MLP model
- 95% Confidence Interval — point estimate $\pm$ RMSE of the trained model
- R^2 Badge — model goodness-of-fit; green ≥ 0.90 , amber ≥ 0.70 , red < 0.70
- Bar visualisation — shows the CI band and point estimate on a 0–200 MPa axis

Concrete Type	RMSE (MPa)	R^2	Data Source
NC	4.35	0.782	Literature + lab data
HPC	5.62	0.739	Literature (limited: 26 rows)
SCC	6.98	0.927	Literature
UHPC	4.30	0.928	Literature
GEO	12.60	0.037	Limited; model under development
HSLWC	5.00	N/A	Local ACI only; no AI model yet

6.3 Carbon Footprint

CO₂ emissions per m³ are calculated using emission factors from the Intergovernmental Panel on Climate Change (IPCC) and published EPDs. The breakdown bar chart shows the relative contribution of each material. OPC dominates due to its high emission factor (0.83 kg CO₂/kg); supplementary cementitious materials significantly reduce the total footprint.

7. History and Comparison Tool

7.1 History Tab

All calculations are automatically saved in the History tab (up to 10 entries; oldest is removed when full). Each history card displays the concrete type, label, calculation time, mode, binder composition summary, AI-predicted f_c , and CO_2 .

Available actions per history item:

- Reload (↺) — restores all inputs to the saved state and displays the saved result
- Compare (□) — marks the entry for side-by-side comparison (max 2 items)
- Delete (🗑️) — permanently removes the entry from the list

7.2 Comparison View

Select exactly two history items using their Compare buttons, then press the 'Compare' button in the blue bar. The comparison table shows:

- AI predicted f_c , CO_2 , W/B ratio, unit weight, total binder content
- Green highlighting indicates the better value for each metric
- A percentage difference badge appears next to the better-performing value
- A secondary table shows the binder composition side-by-side

8. Limitations and Disclaimer

Academic Disclaimer

This application is developed as an academic demonstration for the Application Design Project course at KMUTT. All output values are preliminary estimates intended to guide mix design. A full laboratory trial mix programme must be conducted before any mix is used in structural or critical applications.

- Normal Concrete proportions follow ACI 211.1; aggregate quantities assume SSD condition. Actual specific gravity of local materials may differ.
- GEO (Geopolymer) uses an empirical regression model. Current $R^2 = 0.037$ indicates high uncertainty; additional training data incorporating activator chemistry is required.
- HSLWC uses Local ACI proportioning only. No AI model is available; density values assume expanded clay LWA (apparent $SG \approx 1.4$).
- All AI predictions use an MLP model trained on literature data. Model performance varies by concrete type (see Section 6.2).
- Carbon footprint values are approximate and based on generic emission factors; region-specific EPD data will improve accuracy.

9. Quick Reference Card

Action	How To
Open/close guide	Tap the '?' button in the header
Switch mode	Tap 'Normal' or 'Research' in the mode bar
Change unit (MPa/ksc)	Tap the MPa or ksc toggle button next to f_c input
Add binder (Research)	Tap '+ Add Binder', then tap the material in the picker
Remove binder	Tap the × button on the binder row
Calculate	Tap the orange 'Calculate Mix Proportion' button
Reload history item	Tap '↺ Reload' on the history card
Compare two designs	Tap '☐ Compare' on two cards, then 'Compare →'
Delete history item	Tap the 🗑 button on the history card
Re-open guide	Tap '?' in the header at any time